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Smart Grid Deployment in Archipelagic Systems

Lessons from Indonesia's ArGi Initiative

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Indonesia:
An Archipelagic Power System

Indonesia operates electricity systems across:

- **17,000+ islands**
- **600+ isolated power systems**
- thousands of island grids with very different local condition
- **Geography is the starting point for understanding Indonesia's power system**

The Largest Archipelagic Power System in the World

Many island systems have:

- peak demand **< 10 MW**
- isolated operation
- diesel-based generation

Key message

Indonesia operates **an ecosystem of power systems.**

Why Smart Grid Deployment Must Be Different

Most power system theory is built around continental systems. In these systems, **development starts from a large interconnected grid** and gradually adds distributed resources.

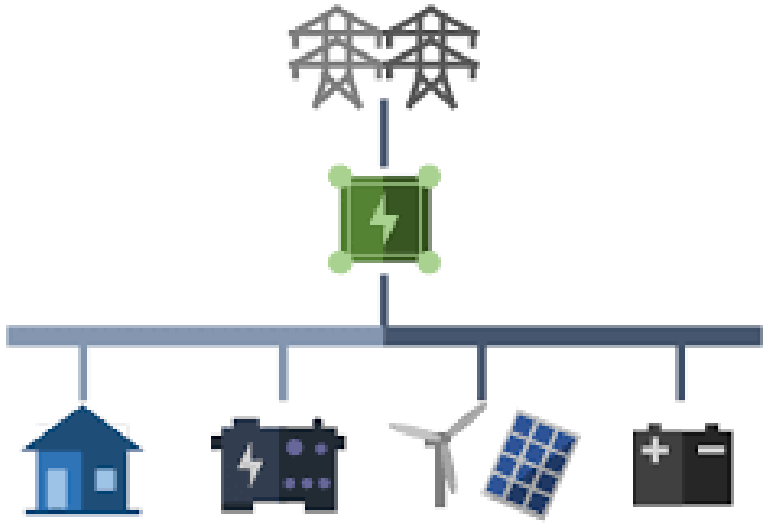
In archipelagic systems, the evolution is the opposite. We **begin with many small microgrids**, then gradually integrate them into larger island systems. This difference fundamentally changes how smart grids must be deployed.

Continental Power System Paradigm

- Large interconnected transmission backbone
- Centralized generation as system anchor
- Distributed energy integrated **after grid maturity**
- Expansion driven by **load growth and grid extension**

Archipelagic Power System Paradigm

- Multiple **isolated island grids**
- Small and medium-scale generation systems
- Renewable microgrids become **initial system backbone**
- Integration occurs through **layered system evolution**



Why This Matters

- Most power system theory assumes **large interconnected grids**
- Planning frameworks prioritize **transmission expansion**
- Operational models assume **centralized system stability**

But **archipelagic systems challenge these assumptions**

They require:

- **distributed-first architecture**
- **microgrid-centric development**
- **evolutionary integration strategies**

Indonesia therefore becomes a **living laboratory for archipelagic power system transformation**.

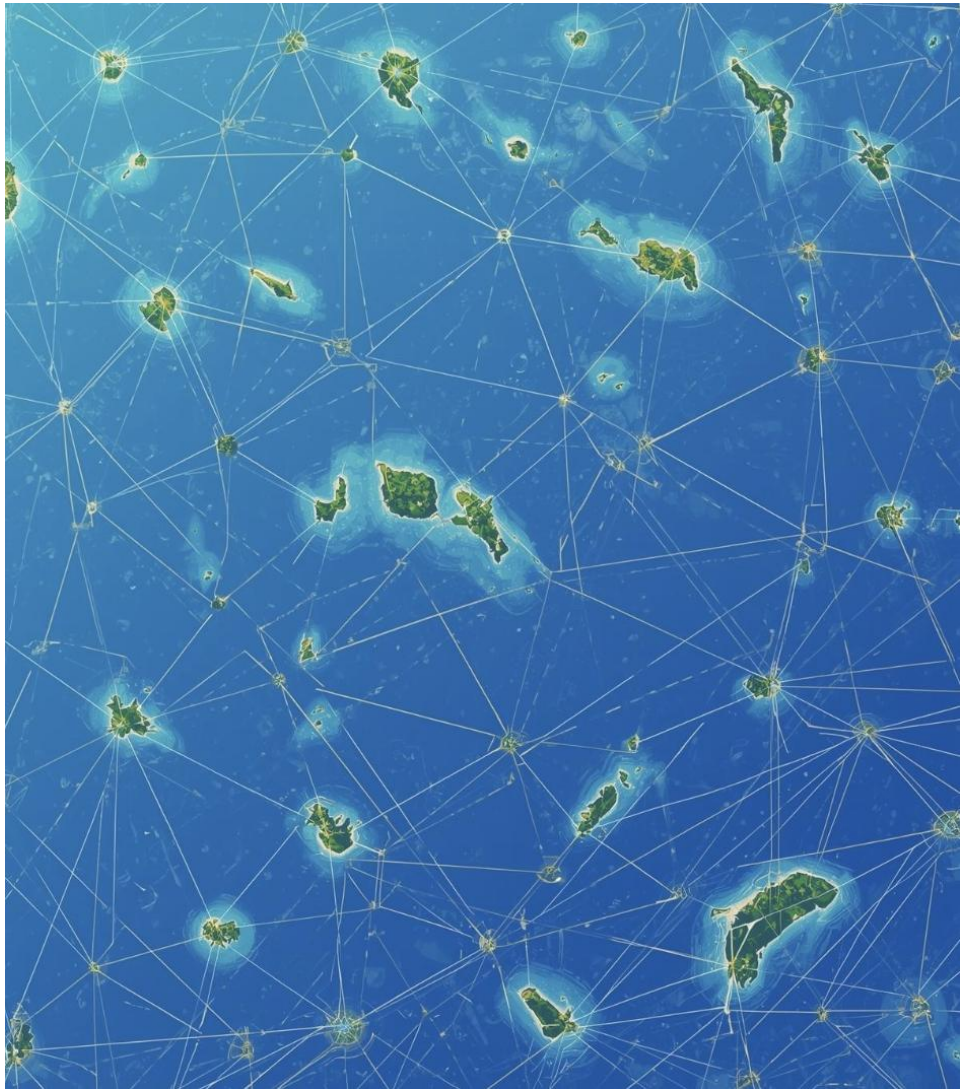
ArGi - Archipelagic Grid Initiative

Indonesia's deployment framework for island power systems.

Principles:

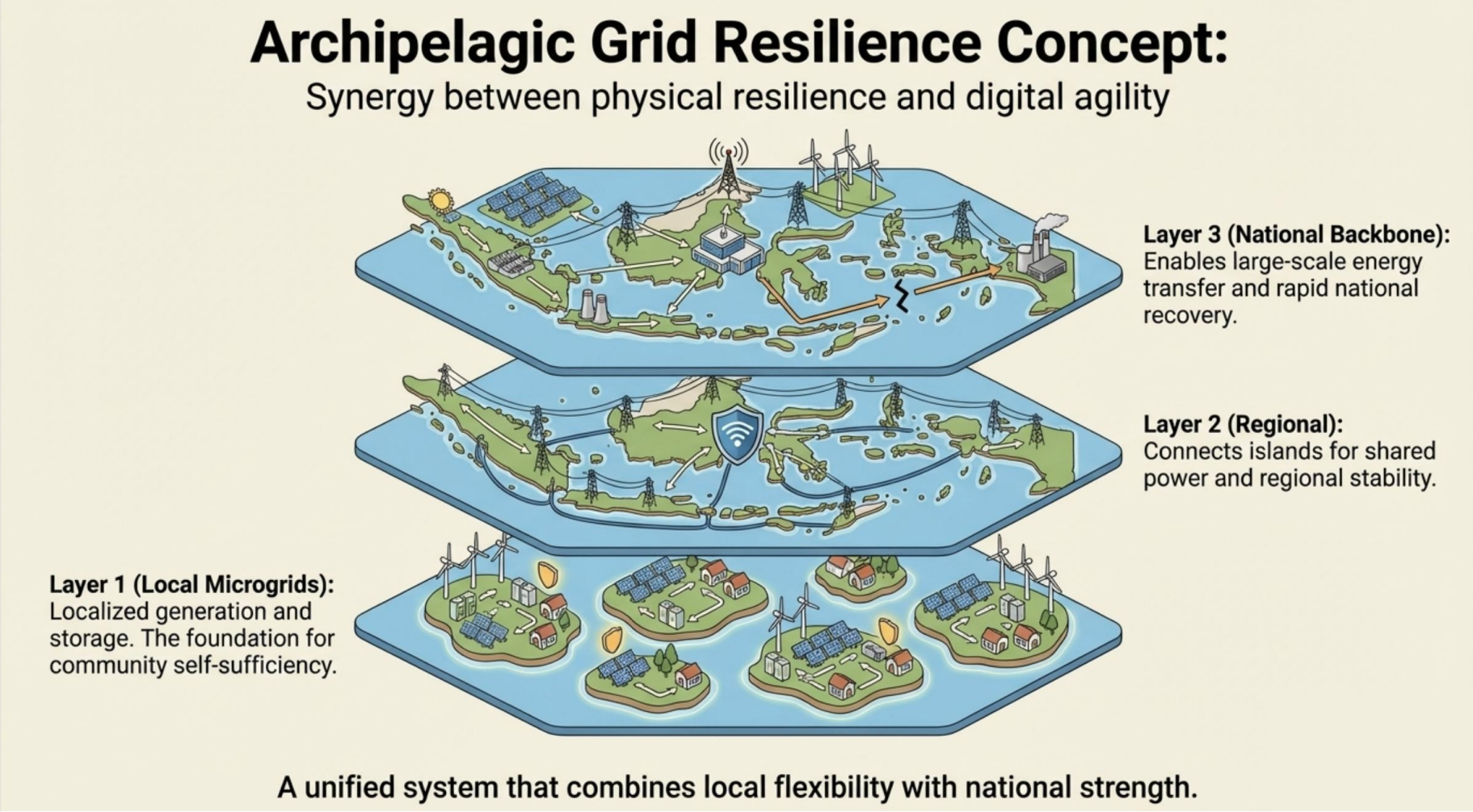
- distributed architecture
- hybrid renewable microgrids
- **staged system evolution**
- sandbox-based learning

ArGi is a staged evolution of archipelagic power systems from micro-islands to interconnected networks.



Layered Architecture for Archipelagic Systems

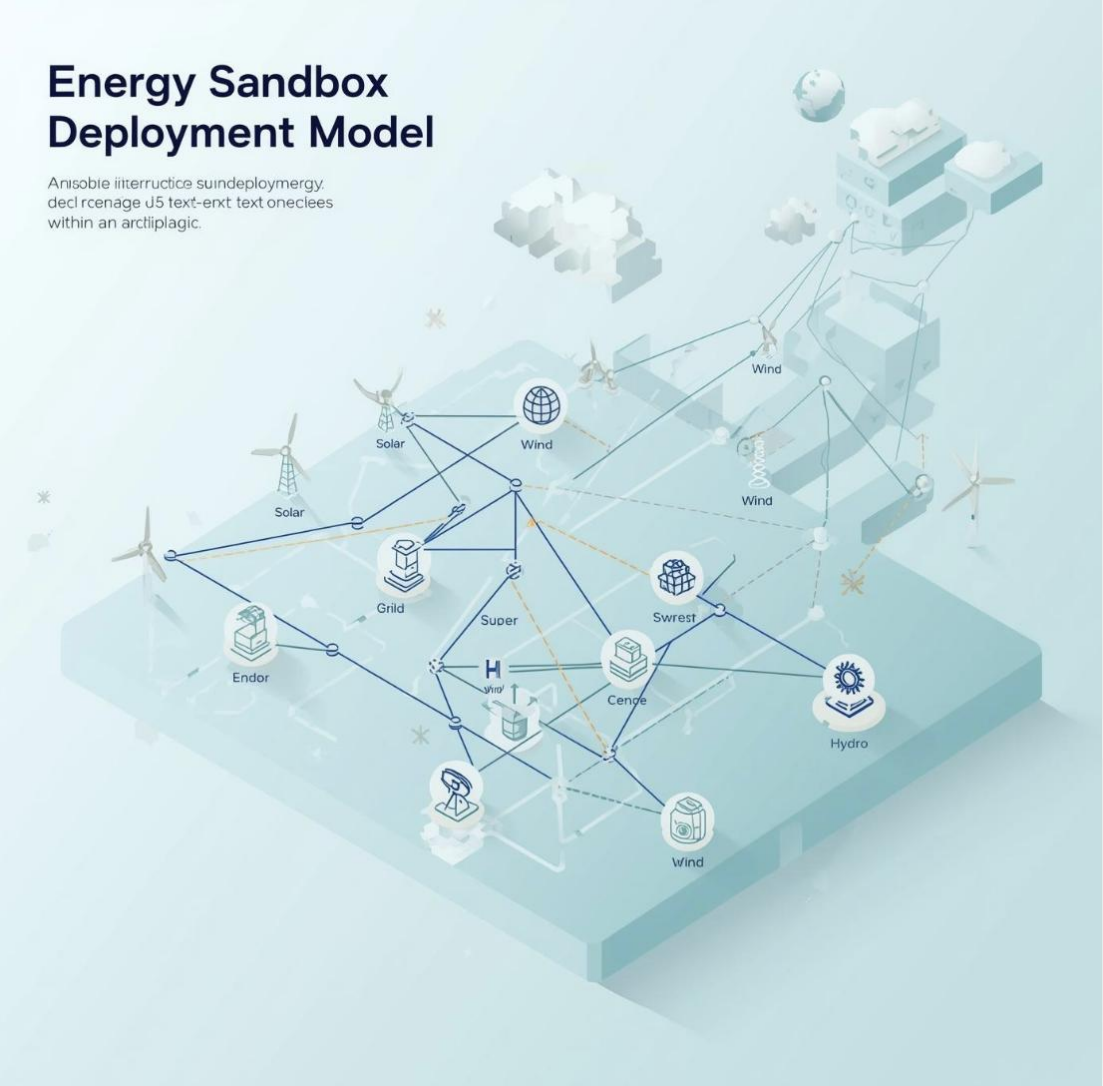
1. Micro-Island Systems
2. Island Grid Systems
3. Archipelagic Network Integration



Energy Sandbox
Deployment Model

Sandbox environments enable:

- real-world experimentation
- technology integration
- **regulatory flexibility**
- operational learning



Technology Colony Ecosystem

Collaborative Innovation for Sustainable Growth

Collaborative Innovation

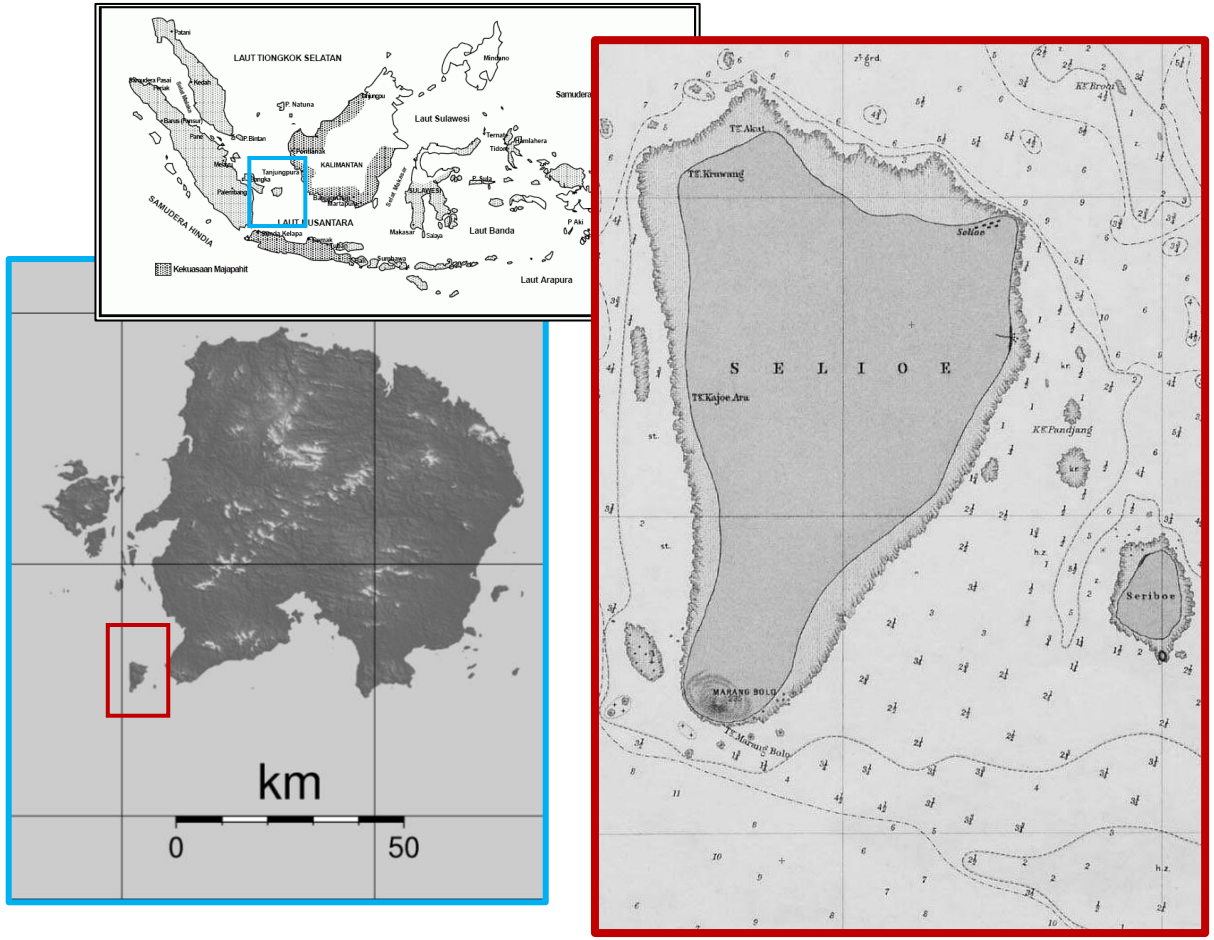
The success of the ArGi initiative relies on **collaborative innovation**, bringing together diverse stakeholders to develop tailored solutions that address unique challenges faced in archipelagic systems.

Technology Colony Ecosystem

Domains:

- Renewable energy systems
- Energy storage technologies
- Smart grid control
- Hydrogen systems
- Digital energy platforms

Seliu Island Microgrid



Pilot Project Overview

Seliu Island serves as a **pioneering pilot microgrid**, demonstrating renewable integration and decentralized energy solutions tailored for Indonesia's unique archipelagic challenges.

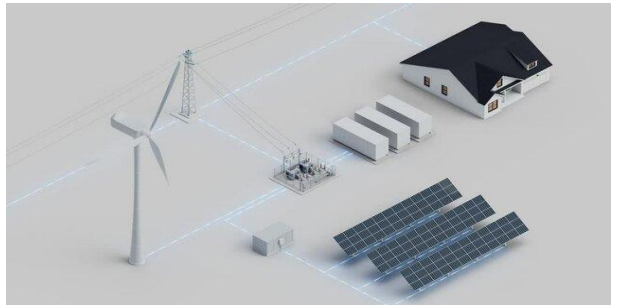
Hybrid Renewable Microgrid

Components:

- solar PV
- battery storage
- microgrid controller
- diesel backup

Solar
Utilizes solar panels for sustainable energy generation.

Battery Storage
Ensures energy reliability and grid stability throughout.



Diesel
Diesel changes role from baseload generation to backup support

Replication Potential

Scalability and Adaptability Insights

Scalability

The ArGi initiative demonstrates that smart grid solutions can be **scaled effectively** across diverse Indonesian islands, leveraging local resources and infrastructure while ensuring sustainable energy access for all communities.

Adaptability

Tailoring the deployment strategy for each island's unique **geographical and cultural context** allows for seamless integration of renewable energy technologies, enhancing resilience and meeting specific community needs efficiently.

Belitung Island

Integrated Energy Sandbox

Existing infrastructure:

- coal power plant **2×16.5 MW**
- diesel generation
- 70 kV network

Integrated Energy Sandbox

Belitung Island serves as a **testbed for innovative energy solutions**, showcasing the integration of renewable resources and smart grid technologies tailored to its unique archipelagic context.

Renewable Integration in Island Grid Systems

Belitung offers renewable opportunities including wind potential and solar development on ex-mining land

- The key challenge is **integrating renewables into an existing fossil-based** island system
- This makes Belitung relevant for studying flexible operation, storage, and smart grid control
- The sandbox focus is transition management, not just renewable installation

Lessons for Archipelagic Nations

Flexibility

Adapting energy solutions to local conditions is crucial for effectively managing diverse island energy demands and resources.

Resilience

Building robust energy systems enhances **capacity to withstand** environmental challenges and ensures reliable power supply even during disruptions.

Indonesia as a Living Laboratory

Scalability

The ArGi initiative demonstrates scalable solutions that can be adapted across various Indonesian islands for smart grid deployment.

Adaptability

Lessons learned from diverse island contexts highlight the adaptability of smart grid technologies to meet unique local challenges.